



Week 07 - Tutorial 01



Transcript Language

English



Hello everyone. In this lecture, in this tutorial video we will talk about similar matrices. So,

let us take an example. So, we start with some linear transformation T of x, y, z

is $x, x + y$ and $x + z$. So, this is a linear transformation, from $\mathbb{R}^3$ to $\mathbb{R}^3$ and now we consider

ordered basis, so this is β_1 , this is the standard ordered basis we are considering $1, 0, 0$;

$0, 1, 0$ and $0, 0, 1$. So, this is the standard ordered basis.



Let us consider another ordered basis β_2 , let us take β_2 as $(1, 0, 1)$;

$(0, 1, 1)$ and $(1, 1, 0)$. So, if we write the matrix representation with respect to both the ordered

basis, so T with respect to β_1 , and for the domain and codomain in both places if we consider

the basis to be β_1 , then this will give us the matrix. So, let us write it T of $(1, 0, 0)$

that is nothing but $(1, 1, 1)$; T of $(0, 1, 0)$ that is the matrix, so $(0, 1, 0)$

and T of $(0, 0, 1)$ this will give us $(0, 0, 1)$.

So, the matrix we got here, so this matrix is nothing but first column will be $(1, 1, 1)$,

$(0, 1, 0)$ and the third column will be $(0, 0, 1)$.

So, this is the matrix representation with respect to standard ordered basis. Now, similarly if we

calculate T of these vectors in β_2 and we write, we then get T of $(1, 0, 1)$ is nothing but

$(1, 1, 2)$. T of $(0, 1, 1)$ is nothing but $(0, 1, 1)$ and T of $(1, 1, 0)$ is $(1, 2, 1)$.

Now, $(1, 1, 2)$ we can write, express in terms in a linear combination of vectors in β_2 .

So, we will get 1 into 1, 0, 1 plus 1 into 0, 1, 1 plus 0 into 1, 1, 0. Similarly, we

can express the other two also, so the matrix we get is T_{β_2} is given by 1, 1, 0; 0, 1, 0 and

0, 1, 1. So, I want all of you to check this representation. You can express all the images

these vectors and these vectors as a linear combination of vectors in β_2

and then write the matrix representation. So, we get T_{β_1} and T_{β_2} ,

those are the matrix representation with respect to β_1 and β_2 respectively.

Now, these two matrix are similar, but for similar matrix we know that there should exist some,

so let us denote this matrix as A for removing the complication of notation and this matrix as B.

So, for as A and B are similar matrix, so there should exist some P such that $P^{-1}BP = A$

is A or even you can write A is nothing but $P^{-1}BP$, sorry B is nothing but $P^{-1}BP$.

So either of these, basically I just change the size, sides of P. So, we want to find this P.

What will be this P? So, what we got till now,

till now we got that if we consider β_1 for both domain and codomain we get the

matrix A as the representation of linear transformation. Similarly, for β_2 ,

we got the matrix B . So, now if we represent this β_2 in terms of β_1 , then we get

the change of basis matrix that will give us the matrix P . So, what we have to do? We have to

do, we have to express the vectors of β_2 in terms of vectors of β_1 .

So, here β_2 is nothing but $1, 0$, β_2 is nothing but $1, 0, 1; 0, 1, 1$ and

$1, 1, 0$. Now we have to express this in terms of β_1 , β_1 is our standard ordered basis.

So, this is $0, 1, 0$ and $0, 0, 1$. So, this is the standard ordered basis. So, we have to express,

we have to express β_2 in terms of

β_1 . So, this is the matrix which we get in this process, we will be called as

change of basis matrix. So, let us try to write $1, 0, 1$ in terms of

standard ordered basis so which is nothing but 1 into the first one $1, 0, 0; 0$ into the second one

0, 1, 0 and 1 into the third one 0, 0, 1. So, similarly for the other two vectors we

can write this expression, so this is nothing but the matrix which we get by taking the vectors of

β_2 as the columns of this matrix. So, the matrix we get is the column of this,

we replace the columns with the vectors in β_2 . So,

this is the matrix we get, this is the change of basis matrix and we denote it by P .

Now, if we calculate P of B we will get 1, 0, 1; 1, 1, 2 and 2, 1,

1. So, this is our P of B . And if we calculate A into P so this is the matrix P into B

and A into P will give us the same matrix 1, 0, 1; 1, 1, 2; 2, 1, 1. I want you to check verify this

two, these two matrices PB and AP . So, what we get here? We get PB is equal to AP . So, B equal to P

inverse AP . So, we can verify that this A and B are really a similar matrix, but P is constructed

by expressing β_2 in terms of β_1 . Now, in the same example we have not changed

the linear transformation and we have also taken the β_2 , the basis

as $1, 0, 1$; $0, 1, 1$ and $1, 1, 0$ as earlier. So, this basis, B matrix we have calculated earlier.

Similarly, if we take the another basis β_3 as $1, 0, 0$; $1, 1, 0$ and $1, 1, 1$

you can calculate the matrix representation, you can verify it that it will be C.

Now, in this case we have also got two matrices B and C by just changing the basis. So,

this B and C will be similar and we want to find, we want to find some matrix Q such that $QCQ^{-1} = B$

inverse will be B^{-1} . That means, C will be $Q^{-1}BQ$. We want to find this Q. Now, how we can find

this Q? We have to express β_3 , the vectors in β_3 in terms of the vectors in β_2 . We have

to find this matrix and that will be our Q. So, let us try to write the first vector $1, 0, 0$

in terms of the vectors of β_2 . So, that is nothing but a into this

plus b into $0, 1, 1$ and c into $1, 1, 0$. So, what we get here? We get $a + b + c$

and this $b + c$ and $a + b$. So, our $a + b$ will be 1 ; $b + c$ will be 0

and $a + c$, sorry the first one is $a + c$, $a + c$ will be 1 , and the last one is $a + b$

b that is 0. So, we got, we get 3 equations and if we solve it we will get a to be half,

b to be minus half and c to be half. So, similarly we can express the other two vectors

which are 1, 1, 0 and 1, 1, 1 in terms of vectors in β_2 and if we write all these

things in the columns, as the coefficients as the columns of a matrix, then we get the matrix Q.

And I want all of you to verify that this is nothing but half, minus half, half which we

have already checked and the next one will be 0, 0, 1 and the last one will be half, half and half.

So, this is our matrix Q. Now if you calculate BQ, you will get half, 0, half;

half, 1, $\frac{3}{2}$ and half, 1, half. So, this is the matrix BQ and which will be same as QC. So, if you

take product of Q and C and B and Q, you will get the same matrix.

So, we will get C equal to $Q^{-1}BQ$, where Q is nothing but the change of basis matrix.

So, how we can get Q? We just express all the vectors in β_3

in terms of the vectors in β_2 . I will write this coefficient, this a, b and c for each case.

So, write a , b and c as the column of the matrix of Q . So, that is how we can calculate

Q and that is how we get, you can verify that two matrices are similar matrix or not. So, whenever

there is a change of basis and we get two representation of a linear transformation

with respect to that basis, then those two matrices are similar matrices. Thank you.